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Published online 13 May 2022 | https://doi.org/10.1007/s40843-022-2045-4
Sci China Mater 2022, 65(10): 2842–2849

Simultaneously enhanced oxidation resistance and mechanical properties in a novel lightweight $Ti_2VZrNb_{0.5}Al_{0.5}$ high-entropy alloy

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ABSTRACT Enhanced oxidation resistance is a primary demand for the application of refractory high-entropy alloys (RHEAs) at elevated temperatures. In this study, Al was added to a Ti_2VZrNb RHEA to partially substitute Nb to improve its oxidation resistance and mechanical properties. The alloy was found to have an increased oxidation resistance by forming a continuous $Al_2O_3 + ZrO_2$ oxide protective surface. At the same time, the room-temperature yield strength was also increased by 66% to 1273 MPa *via* solid solution strengthening. The low atomic mass of Al also helped to reduce the density of the alloy by 8.2% to 5.44 g cm^{-3}. This resulted in a high specific yield strength of 234 MPa cm^3 g^{-1} for the alloy. Meanwhile, the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy also exhibited a high compressive plasticity of >50%. These values are among the best reported so far for RHEAs.

Keywords: high-entropy alloy, mechanical property, oxidation resistance, strength, plasticity

INTRODUCTION

Refractory high-entropy alloys (RHEAs) are a special group of multiconstituent alloys made of some high-melting-temperature elements, such as Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, and W [1,2]. They generally have melting temperatures above 1600°C and are usually body-centered cubic (BCC) in structure [1,3,4]. These alloys are able to exhibit high-temperature strengths comparable to those of Ni-based single-crystal superalloys, making them potential candidates for turbine blade applications. For example, NbMoTaW and NbMoTaWV alloys are able to retain yield strengths above 400 MPa at 1600°C [5], which are higher than those of most Ni-based single-crystal superalloys at similar temperatures.

High oxidation resistance at elevated temperatures and low density are also critical or desirable requirements for the application as turbine blades. In addition, high plasticity is also a high benefit for the application of these alloys [6–9]. However, most RHEAs suffer from poor plasticity, largely due to the inherent nature of their BCC structure [5,10–13]. They are also generally heavy, apparently related to the heavy refractory elements they contain [14,15]. Their oxidation resistance is also generally low because they do not form protective oxide films [16–18]. These drawbacks limit the engineering application of RHEAs. Thus, it has been a goal for material scientists to develop lightweight and oxidation-resistant RHEAs with high plasticity.

To reduce the density of RHEAs, Al is commonly added to alloys [19–22]. In addition to lowering the alloy density, Al also helps to form a protective oxide film on the alloys, improving their high-temperature oxidation resistance [17,23,24]. A caution of using Al is the formation of intermetallic phases with other host elements due to its large negative mixing enthalpies with most metallic refractory elements. This may increase the alloy strength *via* second phase strengthening [25,26] and reduce the alloy plasticity [13,22,27]. For example, the compressive yield strength of a MoNbTaTiV RHEA was increased from 1230 to 1390 MPa by the addition of 16.7% Al, but the compressive plasticity was decreased from 32% to 10% [28]. In this regard, the formation of second phase intermetallics is undesirable in Al-containing RHEAs.

This study attempts to fill this gap by developing a novel single-phase Al-containing RHEA with a low mass density, high yield strength and high plasticity, and good oxidation resistance. This is achieved by using ductile and light base elements, element substitution by Al, and engineering the valence electron concentration for alloy plasticity. The effects of Al on the alloy microstructure, lattice distortion, and oxidation resistance were investigated.

EXPERIMENTAL SECTION

The alloy design strategy is based on the following considerations (Fig. S1a). (1) First-principles calculations indicate that elements in group IV (Ti, Zr, Hf) and group V (V, Nb, Ta) are intrinsically ductile and fail by shear deformation instead of cleavage under tension [29]. Hence, the base elements were selected from these groups for high plasticity. (2) To reduce alloy mass density, heavy elements Ta (ρ_{Ta} = 16.65 g cm^{-3}) and Hf (ρ_{Hf} = 13.31 g cm^{-3}) were eliminated, and Ti (4.54 g cm^{-3}), V(6.11 g cm^{-3}), Zr (6.50 g cm^{-3}), and Nb (8.57 g cm^{-3}) were selected as the principal elements. (3) The alloys have valence electron concentrations (VECs) below 4.6 for high plasticity

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[30,31]. A low VEC lowers the threshold lattice strain for dislocation slip (easier for dislocations to move) [32]. Based on the above considerations, a Ti_2VZrNb alloy was designed. To improve the oxidation resistance and further reduce the density and VEC, a second alloy was also designed by substituting 50% of the Nb content with Al, giving a nominal composition of $Ti_2VZrNb_{0.5}Al_{0.5}$.

The two alloys were fabricated from high-purity (99.9 at% or better) elemental metals by means of vacuum arc melting in a water-cooled copper crucible in an argon atmosphere. The densities of the alloys were measured *via* the Archimedes method in water using a ME204E balance. The samples were 5 mm × 4 mm × 4 mm in dimension. The samples for oxidation testing were approximately 4 mm × 4 mm × 4 mm, and all of their surfaces were ground using SiC metallurgical abrasive papers to 2000 grit size and then ultrasonically cleaned in ethanol. Isothermal oxidation tests were conducted at 873 and 1073 K in air. Sample weight gains were measured to evaluate the extent of oxidation. Cylindrical specimens of Φ 6.0 mm × 9.0 mm were machined from the ingots by means of electrical discharge machining for compression testing. Compression tests were performed using an Instron 8860 mechanical testing machine under an engineering strain rate of 1×10^{-3} s^{-1}. At least three specimens for each alloy were tested to ensure measurement reliability.

The crystalline structures of the alloys were analyzed by X-ray diffraction (XRD) using a Bruker D8 Advance instrument with monochromatic Cu Kα radiation. The microstructures of the alloys were examined in a scanning electron microscope (SEM; Quanta 650, FEI) equipped with energy-dispersive X-ray spectroscopy (EDS), a backscattered electron (BSE), and an electron backscatter diffraction (EBSD) detector. Samples for transmission electron microscopy (TEM; FEI-Titan-G2) analysis were fabricated by means of ion milling using a GATAN PIPS II 695 instrument.

RESULTS

Phase analysis and microstructure

In a multiconstituent HEA system, several empirical criteria have been proposed for obtaining a ductile single-phase BCC structure, i.e., Ω parameter >1.1, atomic size difference (δ) <6.6%, and VEC below 4.6 [30,31,33,34]. These three parameters are expressed below:

$$\begin{cases} \Omega = \dfrac{T_m \Delta S_{mix}}{|\Delta H_{mix}|}, \\ \Delta H_{mix} = 4\sum_{i<j} c_i c_j \Delta H_{ij}, \\ \Delta S_{mix} = -R\sum_{i=1}^{n} c_i \ln c_i, \end{cases} \quad (1)$$

$$\begin{cases} \delta_r = \sqrt{\sum_{i=1}^{n} c_i \left(1 - \dfrac{r_i}{\bar{r}}\right)^2}, \\ \bar{r} = \sum_{i=1}^{n} c_i r_i, \end{cases} \quad (2)$$

$$\mathrm{VEC} = \sum_{i=1}^{n} c_i (\mathrm{VEC})_i, \quad (3)$$

where c_i and c_j are the mole fractions of constituents i and j, ΔH_{ij} is the mixing enthalpy of constituents i and j, R is the gas content, T_m is the melting temperature of the alloy, which can be determined by thermodynamic calculation, as shown in Fig. S1b and c, r_i is the atomic radius of constituent i, $\bar{r}$ is the average radius of the constituents, and $(\mathrm{VEC})_i$ is the valence electron concentration of constituent i. Table S1 presents the atomic radii and VECs of the constituents of the two alloys. Table S2 presents the enthalpy of mixing of the element pairs. Table 1 presents the values of the Ω parameter, δ, and VEC of the two alloys calculated using Equations (1–3). Both alloys meet the empirical criteria for ductile single-phase solid solution formation.

The equilibrium phase mole fraction *versus* temperature of the two alloys was calculated based on the titanium alloy thermodynamic database using JMatPro software [35,36], as presented in Fig. S1b, c. It can be seen that both alloys are expected to be fully in a single-phase BCC structure, which is consistent with our alloy design strategy.

Fig. 1 shows the microstructures of the two alloys. Fig. 1a, b show the XRD patterns of the two alloys. Both alloys are in a single-phase BCC structure in the as-cast state. The lattice constants were determined to be 0.3290 nm for Ti_2VZrNb and 0.3297 nm for $Ti_2VZrNb_{0.5}Al_{0.5}$. Fig. 1c, d show the SEM images of the two alloys. Both alloys show dendrite structures. Fig. 1e, f show the tri-color EBSD orientation maps of the two alloys, which have similar grain morphology and average sizes. Fig. 1g, h show TEM images of the alloys. The microstructure is uniform and free of second phase inclusions. The selected area electron diffraction (SAED) patterns shown in the insets confirm their BCC single phase structure. Table S3 presents the EDS analysis of the compositions of the two alloys.

The density of the Ti_2VZrNb alloy was measured to be 5.89 g cm^{-3}, and that of $Ti_2VZrNb_{0.5}Al_{0.5}$ was 5.44 g cm^{-3}. These values are much lower than those of many high-temperature alloys, such as Inconel 718 (~7.8 g cm^{-3}) [37] and Ni-based single-crystal superalloys (~8.9 g cm^{-3}) [8].

Mechanical properties

Fig. 2a shows the engineering compressive stress-strain curves of the two alloys at room temperature. The Ti_2VZrNb alloy showed a yield strength of 769 MPa, and the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy showed a yield strength of 1273 MPa. Both alloys exhibited large compressive strains of >50%. Fig. 2b presents a comparison of the room-temperature yield strengths and mass densities of a number of RHEAs. Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ are among the few lightweight RHEAs with the lowest mass densities [3,38–40]. Fig. 2c shows a comparison of the specific yield strengths and compressive fracture strains of these RHEAs. The two alloys exhibited the best compressive plasticity of all RHEAs shown.

Table 1 Parameters for forming the single-phase BCC structure for the two alloys

Alloy	ΔH_{mix} (kJ mol^{-1})	ΔS_{mix} (J K^{-1} mol^{-1})	Ω	δ_r (%)	VEC
Ti_2VZrNb	−0.16	11.07	136.51	6.15	4.4
$Ti_2VZrNb_{0.5}Al_{0.5}$	−11.04	12.23	2.19	6.15	4.2

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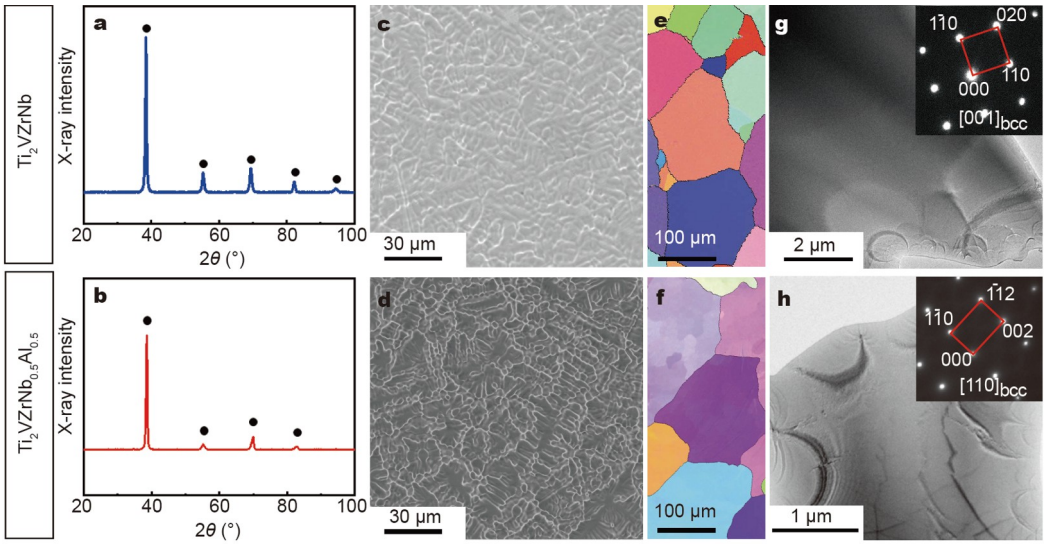


Figure 1 Microstructures of the two alloys. (a, b) XRD patterns confirming the single-phase BCC structure of the two alloys; (c, d) SEM images revealing the dendritic structure of the two alloys; (e, f) tri-color EBSD grain orientation maps of the two alloys; (g, h) bright-field TEM images and corresponding SAED patterns (insets) of the two alloys.

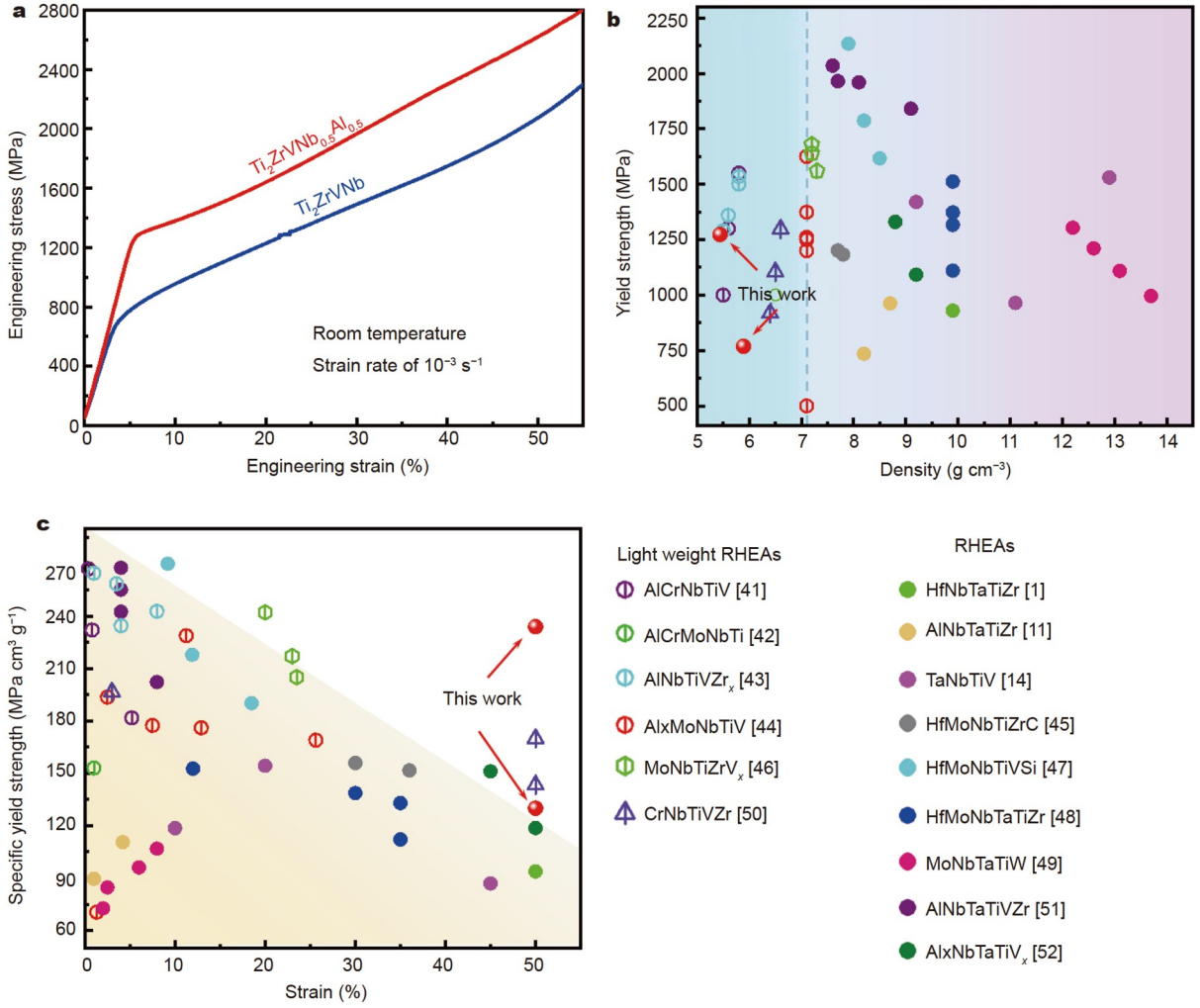


Figure 2 Mechanical properties of the two alloys. (a) Compressive stress-strain curves of the two alloys tested at room temperature; (b) comparison of the mass densities and yield strengths of the two alloys with other RHEAs; (c) comparison of the specific yield strengths and plasticities of the two alloys with other RHEAs [1,11,14,41–52].

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The $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy also exhibited a specific yield strength comparable to that of all the RHEAs. Other Al-containing lightweight RHEAs all show low plasticity. This is caused mostly by the formation of brittle second phases, such as the Laves phase in AlCrNbTaV [41], μ phase in AlCrMoNbTi [42], and Al_3Zr_5 intermetallic phase in AlNbTiVZr [43]. These second phase inclusions promote nucleation of microcracks at the interphase boundaries. The $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy avoided this problem by having a single phase structure.

Oxidation behavior

Fig. S2 presents the oxidation mass gains of the two alloys after thermal exposure in air at 873 and 1073 K. It is seen that $Ti_2VZrNb_{0.5}Al_{0.5}$ experienced a much lower oxidation weight gain than Ti_2VZrNb (Fig. S2a, b). The oxidation resistance benefit of Al addition is significant, especially at lower temperatures. The mass gain of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy after 100 h of exposure at 873 K is only 5.5% of that of the Ti_2VZrNb alloy. Fig. S2c, d show images of the two alloys after 100 h of exposure at 873 K. The Ti_2VZrNb alloy showed severe pest effect damage [17]. The $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy exhibited no such effect, suggesting a protective oxide surface film.

Fig. 3 shows an SEM analysis of the oxide layer formed on the Ti_2VZrNb alloy after thermal exposure. Fig. 3a shows the cross-section of the sample after exposure at 873 K for 10 h. The sample surface exhibited two layers, i.e., oxide layer I and oxygen-ingress layer II, in the base metal above unaffected matrix III. The oxide layer has poor adhesion to the base metal. Fig. 3b shows another sample after exposure at 1073 K for 2 h. It is evident that the thicknesses of both the oxide layer and the oxygen-ingress layer have increased significantly after exposure at 1073 K (2 h) compared with that at 873 K (10 h).

Fig. 3c shows the cross section of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy after exposure at 873 K for 10 h. The oxygen-ingress layer is of similar thickness to that in the Ti_2VZrNb alloy at <10 μm, but the oxide layer is much thinner, approximately 2–3 μm thick. The thin oxide layer also seems to have perfect adhesion to the base metal. Fig. 3d shows the alloy oxidized at 1073 K for 2 h. The inset shows an enlarged view of the oxidized layer. The oxidized surface has two sublayers, i.e., a dense inner oxide layer I and a rough "needle-like" outer layer IV. The dense inner layer is 7–8 μm thick and has perfect adhesion to the base metal. The oxygen ingress layer II has become much thicker than that at 873 K, to ~90 μm. In addition, some image contrasts were observed in layers II and III after exposure at 873 and 1073 K, suggesting some internal phase changes.

Composition analysis was also conducted on the surface layers of the samples. Fig. S3 shows a magnified SEM-BSE image and corresponding EDS point analysis of surface layer I on the Ti_2VZrNb alloy after exposure at 1073 K for 2 h. Four distinctive regions may be recognized based on image contrast. As seen in the data, Region (1) is rich in Zr (brighter contrast), Region (2) is rich in Ti and V (darker contrast), Region (3), the grey matrix background, is rich in Nb and Ti, and the needle-like structure in Region (4) is rich in Zr and Nb (brighter contrast). This observation suggests that the oxidation of the Ti_2VZrNb alloy is nondiscriminatory among the elements and that the oxide formed is a mixture of oxides. Fig. S4 presents an EDS element mapping analysis of the oxides formed on the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy after exposure at 1073 K for 2 h. The rough outer layer (IV) is rich in Al and Zr, and the dense inner layer (I) is rich in Ti. In addition, layer II also displayed clear inhomogeneity for Ti, Zr, V, and Nb. The bright contrast phase appears to be rich in Zr and V, and the dark-contrast phase appears to be rich in Ti and Nb.

Fig. 4 shows an XRD analysis of the oxide scales of the two alloys. Fig. 4a shows the pattern of the Ti_2VZrNb alloy. XRD peaks corresponding to TiO_2, V_2O_5, $TiNb_2O_7$, and $Nb_2Zr_6O_{17}$

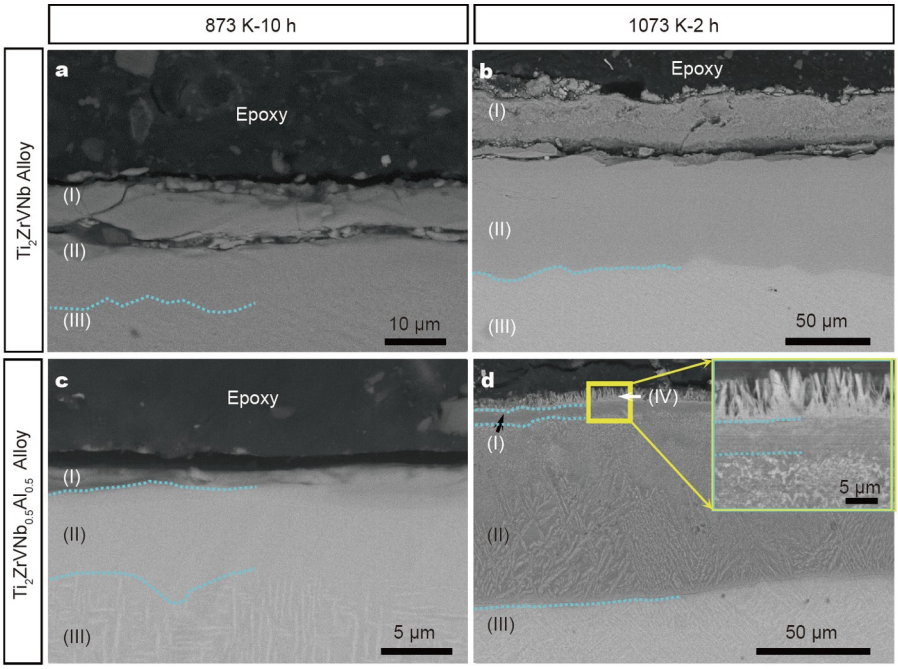


Figure 3 SEM-BSE images of the cross-sectional view of the oxide scales on the Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ alloys. (a) The Ti_2VZrNb alloy oxidized at 873 K for 10 h; (b) Ti_2VZrNb alloy oxidized at 1073 K for 2 h; (c) $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy oxidized at 873 K for 10 h; (d) $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy oxidized at 1073 K for 2 h (the inset is an enlarged image of the oxidized layers).

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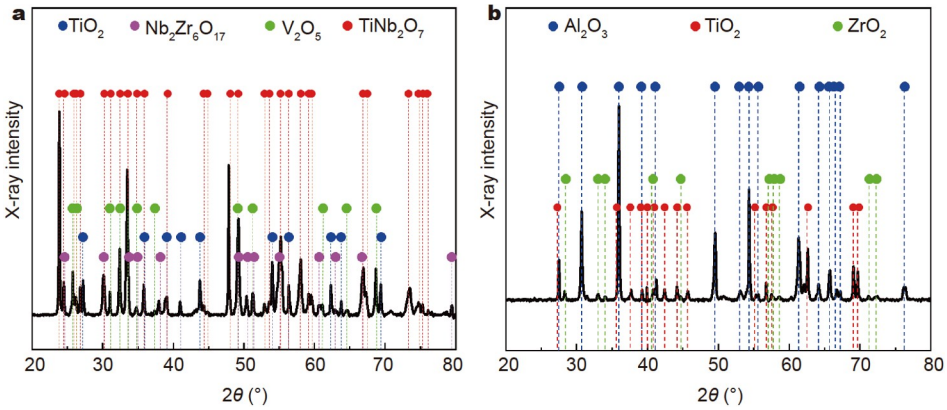


Figure 4 XRD patterns of the surface layers of the two alloys after exposure at 1073 K for 2 h. (a) The Ti_2VZrNb alloy; (b) the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy.

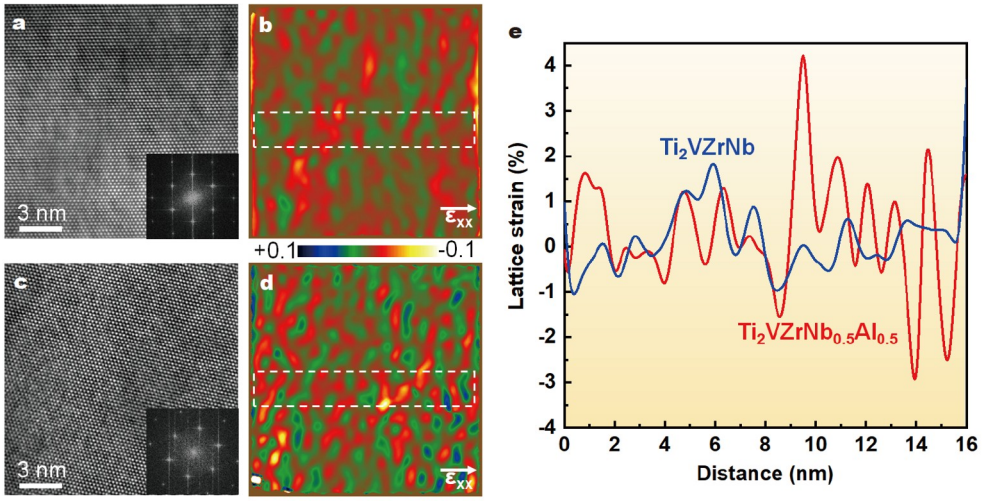


Figure 5 Comparison of lattice strains in the Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ alloys. (a, c) High-resolution HAADF-STEM images of the Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ alloys; (b, d) ε_{xx} lattice strain distribution maps of the viewing areas shown in (a) and (c), respectively; (e) average ε_{xx} lattice strain distribution curves of the two alloys.

can be identified. These observations indicate the formation of a mixture of different oxides. This is also consistent with the SEM and EDS elemental analysis of the presence of all four elements in the oxide layer. Fig. 4b shows the XRD pattern of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy. Most of the major peaks are indexed to Al_2O_3. The presence of TiO_2 is also evident. Some very weak peaks of ZrO_2 may also be recognized. Combined with the EDS analysis of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy (Fig. S4), the outermost rough layer (IV) is identified as Al_2O_3 with a small amount of ZrO_2, and the inner dense layer (I) is identified as TiO_2.

DISCUSSION

The effect of Al on mechanical strength

As presented in Fig. 2, the yield strength of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy is 166% of that of the Ti_2VZrNb alloy. This is attributed to the solid solution strengthening effect. Fig. 5 shows high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of the two alloys in the as-cast state and atomic-scale lattice strain distribution analysis using the peak pairs algorithm (PPA) approach. Fig. 5a, c show the high-resolution HAADF-STEM images of the Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ alloys, and Fig. 5b, d show the lattice strain distribution maps for the horizontal direction (ε_{xx}) of the viewing areas of the Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ alloys. It can be seen that the lattice strain distribution in the Ti_2VZrNb alloy is much more uniform than that in the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy. The average lattice strains for the regions in the dashed boxes (averaged over the seven rows of atoms contained in the boxes) of the two alloys are plotted in Fig. 5e. It is seen that the peak values and the general variations of the lattice strains in the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy are clearly larger than those in the Ti_2VZrNb alloy. The mean square deviation of lattice strains is 1.890 for the $Ti_2VZrNb_{0.5}$ $Al_{0.5}$ alloy and 0.369 for the Ti_2VZrNb alloy. The larger lattice strains in the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy imply a higher resistance to dislocation movement and a stronger solid strengthening effect. In addition, Al is a p-element and is known to form stronger p–d bonds with some refractory d-elements than d–d bonds [12,28,53]. This may further increase lattice friction for dislocation movement.

Effects of Al on oxidation behavior

As presented above, the oxide scale (layer I) formed in the Ti_2VZrNb alloy is a physical mixture of TiO_2, V_2O_5, $TiNb_2O_7$,

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and $Nb_2Zr_6O_{17}$. The Gibbs free energies of formation of the various primary oxides are presented in Fig. S5. It is believed that all four elements are oxidized first to form their primary oxides, possibly due to the lack of a protective layer [18,54,55]. ZrO_2 and TiO_2, however, may further react with Nb to form complex oxides that are more thermodynamically stable, as per [18]:

$$2TiO_2 + 4Nb + 5O_2 = 2TiNb_2O_7, \quad (4)$$

$$12ZrO_2 + 4Nb + 5O_2 = 2Nb_2Zr_6O_{17}. \quad (5)$$

The addition of 10% Al converts the oxide scale into an Al_2O_3 + ZrO_2 outer layer (layer IV) and a TiO_2 inner layer (layer I). Al_2O_3 has a low growth constant [56]. This allows and encourages Zr to oxidize concurrently with Al to form layer IV. The formation of the dense Al_2O_3 + ZrO_2 layer (layer IV) reduces the oxidation of Nb and V. Furthermore, the increased lattice distortion of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy also slows down the inward diffusion of oxygen [57], further reducing the oxidation. This is consistent with the observation of the reduced mass gain rate of the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy shown in Fig. S2b.

The protectiveness of a surface oxide layer against further oxidation is dependent on its physical integrity and its adhesion to the base metal, as well as the Pilling-Bedworth (PB) ratio. Fig. 3 and Fig. S4 show that the oxide scale on the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy is much more intact and has much better adhesion to the base metal, in clear contrast to that on the Ti_2VZrNb alloy. The PB ratios of the various oxides are 1.76 for TiO_2, 6.04 for $Nb_2Zr_6O_{17}$, 1.01 for V_2O_5, and 2.27 for $TiNb_2O_7$. For Ti_2VZrNb, the oxide scale (layer I) contains four main oxides, TiO_2, $Nb_2Zr_6O_{17}$, V_2O_5, and $TiNb_2O_7$. Based on the stoichiometry, the gross PB ratio of the mixed oxides is estimated to be 2.57. This is consistent with the observation of extensive spalling and poor adhesion to the base metal observed in the Ti_2VZrNb sample. For the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy, the addition of Al converted the surface oxide scale to an outer Al_2O_3 + ZrO_2 layer and an inner dense TiO_2 layer. The PB ratios of Al_2O_3 and ZrO_2 were 1.28 and 1.56, respectively. Hence, the mixed oxides for layer IV show a desirable PB ratio in the range of 1–2 [24], which is consistent with the observation of continuous and well-adhered layer IV.

CONCLUSIONS

A novel lightweight RHEA was designed in this study to achieve a combination of high specific yield strength, high room-temperature compressive plasticity, and good oxidation resistance. The main conclusions are summarized as follows.

(1) Two new lightweight refractory high-entropy alloys, Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$, are designed. Both alloys are in the single BCC phase.

(2) The Ti_2VZrNb and $Ti_2VZrNb_{0.5}Al_{0.5}$ alloys exhibit over 50% room-temperature compressive plasticity. The 50% substitution of the Nb content in the alloy by Al reduces the mass densities from 5.89 to 5.44 g cm^{-3} and enhances the room-temperature yield strength from 769 to 1273 MPa. These results translate into an increase in the specific yield strength from 130 to 234 MPa cm^3 g^{-1}.

(3) The Ti_2VZrNb alloy was found to oxidize uniformly among its constituent elements with a significant rate of mass gain. The mixed oxide scale experiences excessive spallation and has poor adhesion to the base metal. The oxides provide no protection to the metal. With Al addition, the mixed oxide scale is changed into a dual-layer structure consisting of an Al_2O_3 + ZrO_2 outer layer and a dense TiO_2 inner layer for the $Ti_2VZrNb_{0.5}Al_{0.5}$ alloy. This oxide scale significantly reduces the oxidation rate of the alloy.

Received 16 January 2022; accepted 21 March 2022; published online 13 May 2022

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Acknowledgements This research was supported by the National Key R&D Program of China (2021YFA1200201), the National Natural Science Foundation of China (52071003, 91860202, and 11604006), Beijing Nova Program (Z211100002121170), Beijing Municipal Education Commission Project (PXM2020_014204_000021 and PXM2019_014204_500032), Beijing Outstanding Young Scientists Projects (BJJWZYJH01201910005018), Beijing Natural Science Foundation (Z180014), and "111" project (DB18015).

Author contributions Han X, Mao S, and An Z designed the study. An Z, Yang T, Yang X, Wang X, and Liu S carried out the main experiments. An Z, Han X, Mao S, Zhang Z, Liu Y, and Chen Y analyzed the data and wrote the main draft of the paper. All authors contributed to the discussion of the results and commented on the manuscript.

Conflict of interest The authors declare that they have no conflict of interest.

Supplementary information Supporting data are available in the online version of the paper.



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协同提高新型轻质$Ti_2VZrNb_{0.5}Al_{0.5}$高熵合金抗氧化性能与力学性能

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摘要 提高抗氧化性能是难熔高熵合金高温应用的首要要求. 本研究中, 我们将Al代替50%Nb添加到Ti_2VZrNb难熔高熵合金中, 以提高其抗氧化性能与力学性能. 结果表明, 相比于Ti_2VZrNb合金, $Ti_2VZrNb_{0.5}Al_{0.5}$合金的抗氧化性得到了显著提升, 这是由于在合金表面形成了连续的Al_2O_3 + ZrO_2氧化层保护膜. 同时, $Ti_2VZrNb_{0.5}Al_{0.5}$合金室温屈服强度高达1273 MPa, 相比于Ti_2VZrNb合金强度提升66%. 结构表征说明, $Ti_2VZrNb_{0.5}Al_{0.5}$合金强度提升得益于其较大的固溶强化效果. 较轻元素Al的添加使得该合金具有较低的密度(5.44 g cm^{-3}), 相比于Ti_2VZrNb合金密度降低8.2%, 这使得该合金的比强度高达234 MPa cm^3 g^{-1}. 同时, $Ti_2VZrNb_{0.5}Al_{0.5}$合金也具有较高的压缩塑性(>50%). 综合这些结果, 该合金是迄今为止所报道的性能最好的难熔高熵合金.

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